PORTFOLIO RISK SUMMARY

Equal-weighted 18-asset portfolio (15 technology-sector equities + BTC, ETH, SOL) · Daily data through October 12, 2026 · Yahoo Finance / FRED · K-Means volatility regimes

30-DAY VOL (ANN.)	**20.9%**	vs 51% in crisis regime
1-DAY 95% VAR	**2.5%**	99% VaR 3.6%
MAX DRAWDOWN	**-53.3%**	Nov 2021 → Dec 2022
AVG PAIRWISE CORR	**0.21**	vs 0.31 in low regime
CURRENT REGIME	**Low**	57 trading days
95% VAR BREACH RATE	**4.2%**	42 of 1004 days vs 5% expected

FINDINGS

- **Diversification broke down under stress.** Average pairwise correlation across the 18-asset book was **0.31** in the low-vol regime but **0.58** in the crisis regime; the equity–crypto correlation rose from 0.17 to 0.44. The crypto sleeve (71% avg annualized vol vs 43% for the equity sleeve) moved with — not against — equity stress.
- **Volatility is regime-structured, not continuous.** K-Means clustering on rolling volatility, correlation dispersion and VIX separated the period into 4 regimes; 10% of days were crisis-regime conditions, the longest spell lasting 57 trading days (May 2022–Aug 2022). Portfolio vol averaged 22% in the calm regime versus 51% in crisis — a 2.4× step-up when the regime flips.
- **Historical VaR is a floor, not a ceiling.** The trailing 1-year 95% historical VaR stands at **2.5%** (99%: 3.6%). In backtest the 95% level was breached 42 of 1004 days (4.2% vs 5% expected), and the worst single day (-6.4%) was 2.5× the current 95% VaR. Breaches cluster around regime transitions, when the 250-day estimation window still carries the calm regime's distribution.
- **Drawdown risk is dominated by regime episodes.** The equal-weighted portfolio's maximum drawdown was **-53.3%** (Nov 2021 peak → Dec 2022 trough, recovering 344 days later); the deepest single-name drawdown was SOL-USD at -96.1%. Current regime: **Low** for 57 trading days, with 30-day vol at 20.9% and average pairwise correlation at 0.21.

RECOMMENDATIONS

1. **Size to the stress regime, not the current one.** Set position limits and volatility targets off crisis-regime vol (51% portfolio, 0.58 avg correlation) rather than the current 30-day vol (20.9%) — otherwise the book is structurally oversized the moment the regime shifts.
2. **Stress-test at crisis correlations.** Use the crisis-regime correlation matrix (avg pairwise 0.58; equity–crypto 0.44) for risk limits and hedge sizing. Name-level diversification provides materially less protection than the full-sample matrix implies; sleeve-level hedges (index puts, futures, cash) are the reliable lever.
3. **Upgrade the VaR process.** Report 99% VaR (3.6%) and expected shortfall alongside the 95% headline, re-estimate on every regime transition rather than on a fixed calendar, and treat breach clusters (4.2% realized vs 5% expected) as a model-risk KPI reviewed monthly.
4. **Operationalize the regime flag.** Codify de-risking triggers on transitions into Elevated/Crisis states (average episode 31 trading days, max 57) with a pre-agreed playbook per regime — the value is in the speed of the response, not the label itself.

Methodology. Daily total returns from adjusted closes (Yahoo Finance); macro context from FRED (10Y yield, VIX, broad dollar index). Volatility annualized on a 252-day basis; historical VaR = 5th/1st percentile of trailing 250-day daily returns (percentile method; no parametric or Monte Carlo assumptions); drawdown on the equal-weighted, daily-rebalanced index. Regimes: K-Means (k=4, fixed seed) on standardized 30-day average volatility, cross-asset vol dispersion, 90-day average pairwise correlation, portfolio vol and VIX; labels smoothed with a 5-day majority filter for episode reporting. **Limitations.** Historical VaR assumes the recent past is representative of the near future — it is not a complete risk model; correlations and volatilities are estimates and cluster around regime transitions; the crypto sleeve is mapped onto the US equity calendar (weekend moves book to the next trading day); equal-weight daily rebalancing ignores transaction costs and taxes. This memo is an analytics exercise, not investment advice.